



Code Transformation Summary by Amazon Q

Q has made the following changes during the transformation process.

Lines of code in your application: 3175

Transformation duration: 13 min(s)

Dependencies upgraded: 9 out of 9

There are no deprecated APIs detected in this app

Files changed: 35

Build status in Java 21: **SUCCEEDED**

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Build log summary

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Amazon Q successfully built the upgraded code in Java 21. Here is a relevant snippet from the build log. To view the full build log, open `buildCommandOutput.log`

```
The build process was successful, completing in under 3 seconds. The project was compiled using Java 21, with both main and test classes being processed. Some unchecked or unsafe operations were detected during compilation, but did not prevent the build from succeeding.
```

Q Generated Summary and Recommendations

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Summary

Q has made the following changes during the transformation process. Please note that the summary is generated by GenAI.

1. **JUnit 5 Migration:** Multiple test files have been updated to use JUnit 5. The import statement has changed from `import org.junit.Test;` to `import org.junit.jupiter.api.Test;`. This change affects numerous files, including `ResourceAndResourceLoaderTest.java`, `LazyInitTest.java`, `FactoryBeanTest.java`, `PackageScanTest.java`, `AwareInterfaceTest.java`, `AutoProxyTest.java`, `ApplicationContextTest.java`,

EventAndEventListenerTest.java, TypeConversionFirstPartTest.java, ValueAnnotationTest.java, PrototypeBeanTest.java, and DynamicProxyTest.java. Additionally, in DynamicProxyTest.java, `@Before` has been replaced with `@BeforeEach`. This upgrade brings improved test discovery, more powerful assertions, and better support for Java 8+ features.

2. **Pattern Matching for instanceof:** Several files have been updated to use pattern matching for instanceof, a feature introduced in Java 14. For example:

- In AbstractAutowireCapableBeanFactory.java:

```
if (beanPostProcessor instanceof InstantiationAwareBeanPostProcessor
processor) {
    Object result = processor.postProcessBeforeInstantiation(beanClass,
beanName);
}
```

- In DefaultAdvisorChainFactory.java:

```
if (advisor instanceof PointcutAdvisor pointcutAdvisor) {
    if
(pointcutAdvisor.getPointcut().getClassFilter().matches(actualClass)) {
        MethodMatcher mm =
pointcutAdvisor.getPointcut().getMethodMatcher();
        // ...
    }
}
```

This change simplifies the code, reduces the risk of ClassCastException, and improves readability.

3. **Dependency Updates:** The `pom.xml` file has been updated with several dependency changes:

- AspectJ Weaver updated to version `1.9.22.1`
- Dom4j updated to version `2.1.4`
- Spring AOP added with version `6.1.4`
- Hutool All updated to version `5.8.36`
- AssertJ Core updated to version `3.27.3`
- JUnit Jupiter added with version `5.12.2`
- Removed dependencies: aopalliance, cglib, and JUnit 4

4. **Text Block Usage:** In ConversionServiceFactoryBean.java, a multi-line string has been replaced with a text block:

```
throw new IllegalArgumentException("""
    Each converter object must implement one of the \
    Converter, ConverterFactory, or GenericConverter interfaces\
    """);
```

This change improves code readability and maintainability for multi-line strings.

5. **URL Creation:** In DefaultResourceLoader.java, the method for creating a URL has been updated to use URI:

```
URL url = URI.create(location).toURL();
```

This change provides a more robust way of creating URLs, handling special characters better.

6. **List API Update:** In `DefaultListableBeanFactory.java`, the code has been updated to use the new `getFirst()` method instead of `get(0)`:

```
return getBean(beanNames.getFirst(), requiredType);
```

This change utilizes the new API introduced in Java 21 for the `List` interface, providing a more semantically clear way to access the first element of a list.

7. **Enhanced Type Inference:** In `PropertyPlaceholderConfigurer.java`, the code has been updated to use enhanced type inference:

```
if (value instanceof String string) {  
    value = resolvePlaceholder(string, properties);  
}
```

This change improves code readability and reduces the need for explicit casting, leveraging Java 14+ pattern matching features.

8. **Simplified Aware Interface Handling:** In `ApplicationContextAwareProcessor.java`, the code has been updated to use pattern matching for instanceof:

```
if (bean instanceof ApplicationContextAware aware) {  
    aware.setApplicationContext(applicationContext);  
}
```

This change simplifies the handling of Aware interfaces, improving code readability and reducing the risk of errors.

9. **BeanReference Handling:** In `AbstractAutowireCapableBeanFactory.java`, the handling of `BeanReference` has been simplified using pattern matching:

```
if (value instanceof BeanReference beanReference) {  
    value = getBean(beanReference.getBeanName());  
}
```

This change improves code readability and reduces the need for explicit casting.

10. **Pattern Matching in AbstractApplicationContext:** The `AbstractApplicationContext.java` file has been updated to use pattern matching for instanceof:

```
if (conversionService instanceof ConversionService service) {  
    beanFactory.setConversionService(service);  
}
```

This change simplifies the code and improves readability.

Recommendations

Q was able to compile the upgraded app successfully. Here are detailed recommendations to further improve the upgrade. Please note that these recommendations are generated by GenAI, apply them accordingly.

- 1. Please review and accept the code changes using the diff viewer. If you are using a Private Repository, please ensure that updated dependencies are available.
- 2. If this project uses `Maven CheckStyle, Enforcer, FindBugs` or `SpotBugs plugins`, Q disabled these plugins when building the project to verify proposed upgrades. Please enable them and verify the code again.
- 3. To help transform the rest of your code, you can use Amazon Q chat in the IDE. You can ask Amazon Q to review the partially updated files and provide new code to address issues, such as compilation errors. You can also use features like `/dev` and `@workspace` to include more of your project as context and get suggestions for multiple files at a time.

Planned dependencies replaced

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Amazon Q updated the following dependencies that it identified in the transformation plan

Dependency	Action	Previous version in Java 8	Current version in Java 21
<code>aopalliance:aopalliance</code>	Removed	1.0	-
<code>cglib:cglib</code>	Removed	3.3.0	-
<code>junit:junit</code>	Removed	4.11	-

Additional dependencies added

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Amazon Q updated the following additional dependencies during the upgrade

Dependency	Action	Previous version in Java 8	Current version in Java 21
<code>cn.hutool:hutool-all</code>	Updated	5.5.0	5.8.36
<code>org.aspectj:aspectjweaver</code>	Updated	1.8.0	1.9.22.1
<code>org.assertj:assertj-core</code>	Updated	3.11.1	3.27.3
<code>org.dom4j:dom4j</code>	Updated	2.1.3	2.1.4
<code>org.junit.jupiter:junit-jupiter</code>	Added	-	5.12.2
<code>org.springframework:spring-aop</code>	Added	-	6.1.4

Deprecated code replaced

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Amazon Q replaced the following instances of deprecated code. An instance with 0 files changed indicates Amazon Q wasn't able to replace the deprecated code.

Deprecated code	Files changed
Other changes	
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All files changed	
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File	Action
pom.xml	Updated
src/main/java/org/springframework/aop/framework/DefaultAdvisorChainFactory.java	Updated
src/main/java/org/springframework/beans/factory/PropertyPlaceholderConfigurer.java	Updated
src/main/java/org/springframework/beans/factory/support/AbstractAutowireCapableBeanFactory.java	Updated
src/main/java/org/springframework/beans/factory/support/AbstractBeanFactory.java	Updated
src/main/java/org/springframework/beans/factory/support/DefaultListableBeanFactory.java	Updated
src/main/java/org/springframework/beans/factory/support/DisposableBeanAdapter.java	Updated
src/main/java/org/springframework/context/support/AbstractApplicationContext.java	Updated
src/main/java/org/springframework/context/support/ApplicationContextAwareProcessor.java	Updated
src/main/java/org/springframework/context/support/ConversionServiceFactoryBean.java	Updated
src/main/java/org/springframework/core/io/DefaultResourceLoader.java	Updated
src/test/java/org/springframework/test/aop/AutoProxyTest.java	Updated
src/test/java/org/springframework/test/aop/DynamicProxyTest.java	Updated
src/test/java/org/springframework/test/aop/PointcutExpressionTest.java	Updated
src/test/java/org/springframework/test/aop/ProxyFactoryTest.java	Updated
src/test/java/org/springframework/test/expanding/PropertyPlaceholderConfigurerTest.java	Updated
src/test/java/org/springframework/test/ioc/ApplicationContextTest.java	Updated
src/test/java/org/springframework/test/ioc/AutowiredAnnotationTest.java	Updated
src/test/java/org/springframework/test/ioc/AwareInterfaceTest.java	Updated
src/test/java/org/springframework/test/ioc/BeanDefinitionAndBeanDefinitionRegistryTest.java	Updated
src/test/java/org/springframework/test/ioc/BeanFactoryPostProcessorAndBeanPostProcessorTest.java	Updated
src/test/java/org/springframework/test/ioc/CircularReferenceWithProxyBeanTest.java	Updated
src/test/java/org/springframework/test/ioc/CircularReferenceWithoutProxyBeanTest.java	Updated
src/test/java/org/springframework/test/ioc/EventAndEventListenerTest.java	Updated
src/test/java/org/springframework/test/ioc/FactoryBeanTest.java	Updated

File	Action
src/test/java/org/springframework/test/ioc/InitAndDestoryMethodTest.java	Updated
src/test/java/org/springframework/test/ioc/LazyInitTest.java	Updated
src/test/java/org/springframework/test/ioc/PackageScanTest.java	Updated
src/test/java/org/springframework/test/ioc/PopulateBeanWithPropertyValuesTest.java	Updated
src/test/java/org/springframework/test/ioc/PrototypeBeanTest.java	Updated
src/test/java/org/springframework/test/ioc/ResourceAndResourceLoaderTest.java	Updated
src/test/java/org/springframework/test/ioc/TypeConversionFirstPartTest.java	Updated
src/test/java/org/springframework/test/ioc/TypeConversionSecondPartTest.java	Updated
src/test/java/org/springframework/test/ioc/ValueAnnotationTest.java	Updated
src/test/java/org/springframework/test/ioc/XmlFileDefineBeanTest.java	Updated